



Mechanical Properties I

Outline

- Introduction
- Mechanical testing
- Tensile test
- Elastic deformation



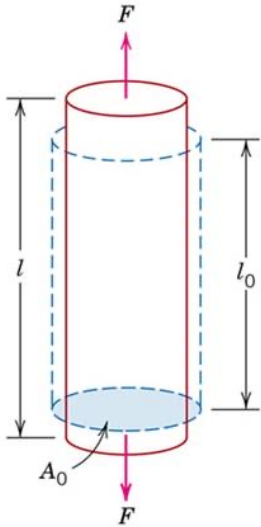
Mechanical Properties

- From an applications standpoint, one of the most important topics within Materials Sc. & Eng. is the study of how materials respond to external loading or deformation.
- Most components, even if used primarily for other property (e.g. electronic substrate) have to fulfil certain *mechanical functions* as well.
- Important mechanical properties are strength, hardness, stiffness, ductility and toughness.
- Laboratory testing to measure mechanical properties attempts to replicate the service conditions.
- Consistency is accomplished by using standardised test, so people are measuring same thing in the same way.
- **American Society for Testing Materials (ASTM)** maintains and updates standards for mechanical properties.
- Several other standards organizations exist, e.g. SAE, ANSI, DIN....

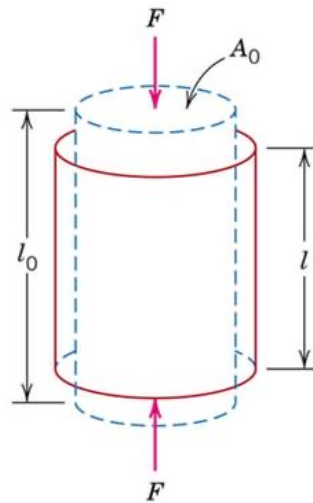


Mechanical Testing of Metals

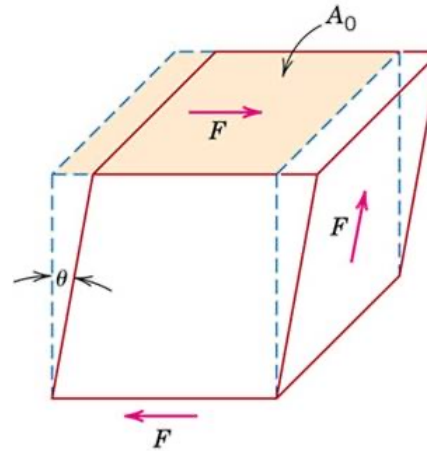
Loading can take any of the following forms:



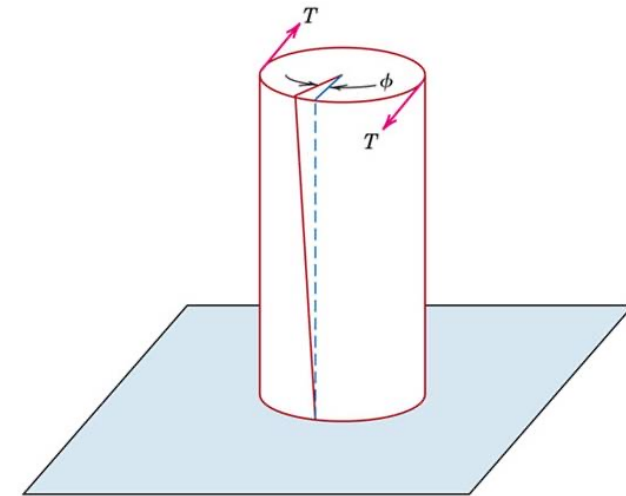
*Pulling/
Stretching*



*squeezing/
squashing*



sliding



twisting

Different tests measure different types of loading conditions



Common States of Stress

Simple tension: cable



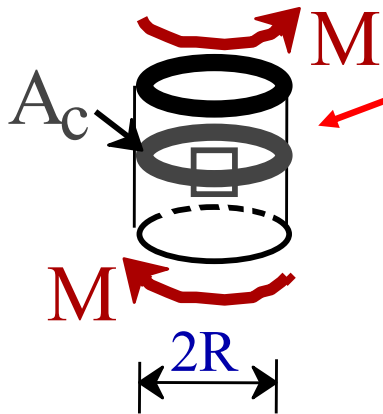
A_0 = cross sectional area (when unloaded)

$$\sigma = \frac{F}{A_0}$$



Ski lift (photo courtesy P.M. Anderson)

Torsion: drive shaft



Simple compression:

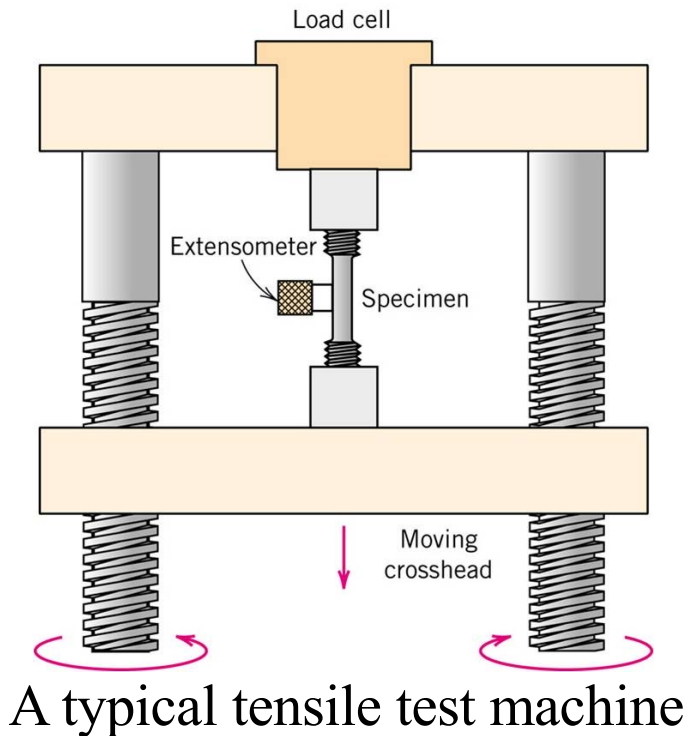


Canyon Bridge, Los Alamos, NM
(photo courtesy P.M. Anderson)



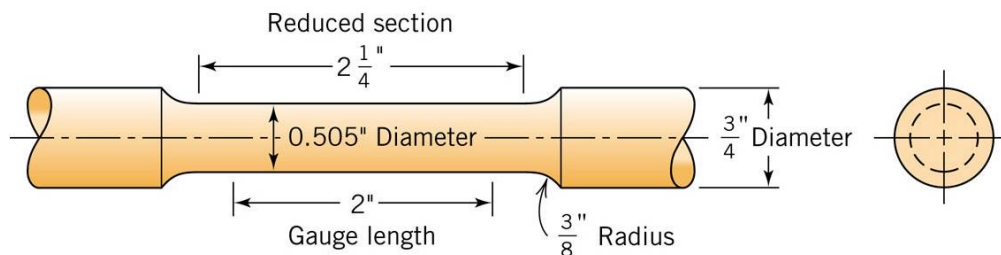
Tensile Test

- The material's response to the applied tensile or compressive load is a change in length.
- We can monitor the change in length very precisely with an instrument called an *extensometer*.



$$\varepsilon = \frac{l_i - l_o}{l_o} = \frac{\Delta l}{l_o}$$

We call this quantity strain



A typical standard tensile test specimen

Where:

l = instantaneous length

l_o = initial length

Strain is a dimensionless quantity (or, can be reported as m/m or in./in.)



Engineering Strain

- Tensile strain:

$$\epsilon = \frac{\delta}{L_o}$$

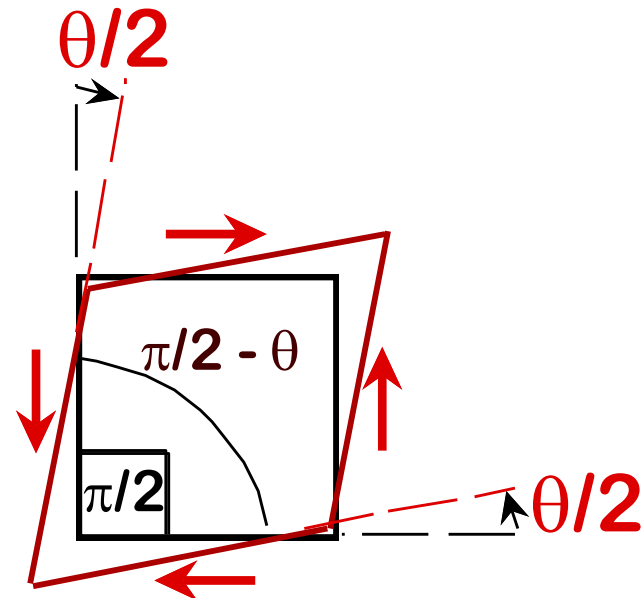
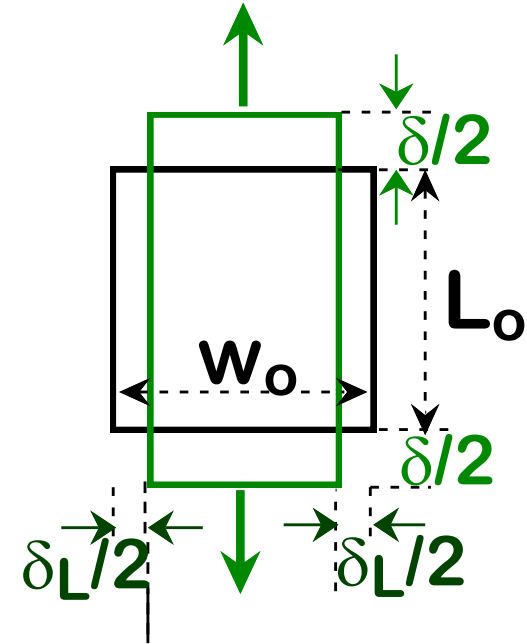
- Lateral strain:

$$\epsilon_L = \frac{-\delta_L}{w_o}$$

- Shear strain:

$$\gamma = \tan \theta$$

Strain is always dimensionless.





Tensile Test

Typically, loading is normalized to cross sectional area:

$$\sigma = \frac{F}{A_o}$$

We refer to this ratio as the applied **stress** when normalized to initial area, this is **engineering stress**.

when normalized to actual area, this is **true stress**.

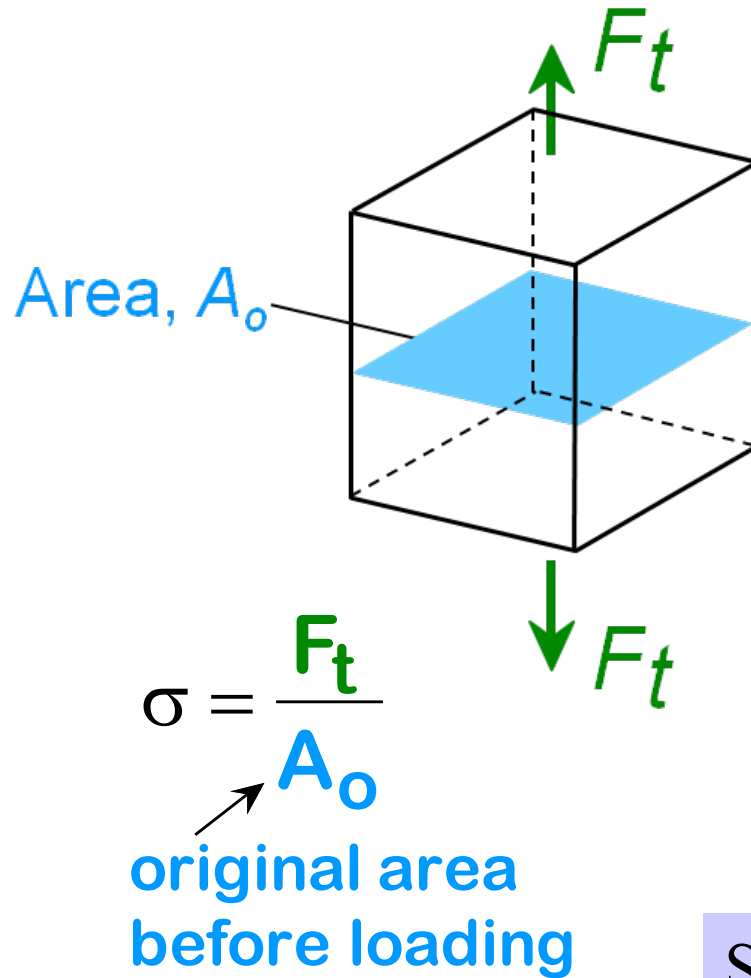
Tensile test

- Load - elongation testing
- But a thin wire breaks under a lower tensile load than a thicker wire of the same material
- Engineering stress (σ) versus engineering strain (ϵ)

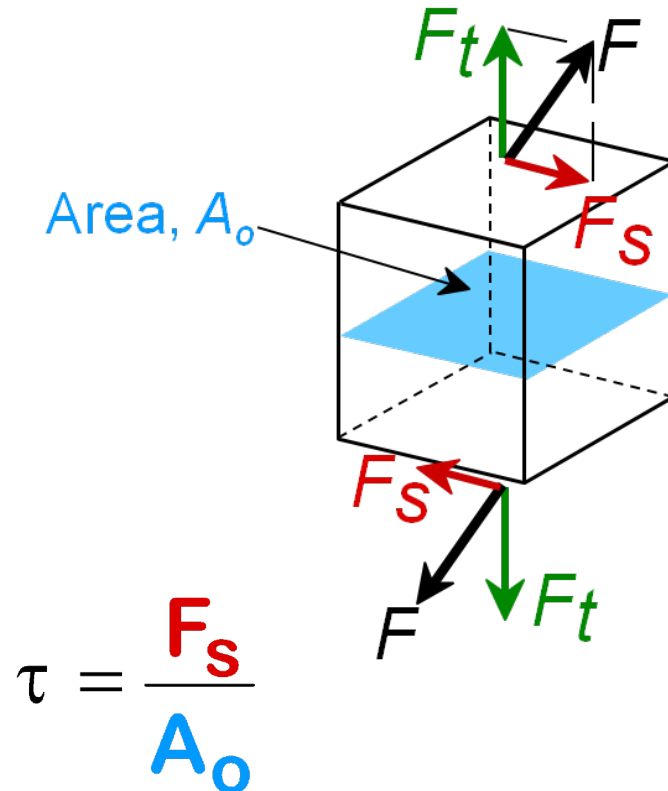


Engineering Stress

- Tensile stress, σ :



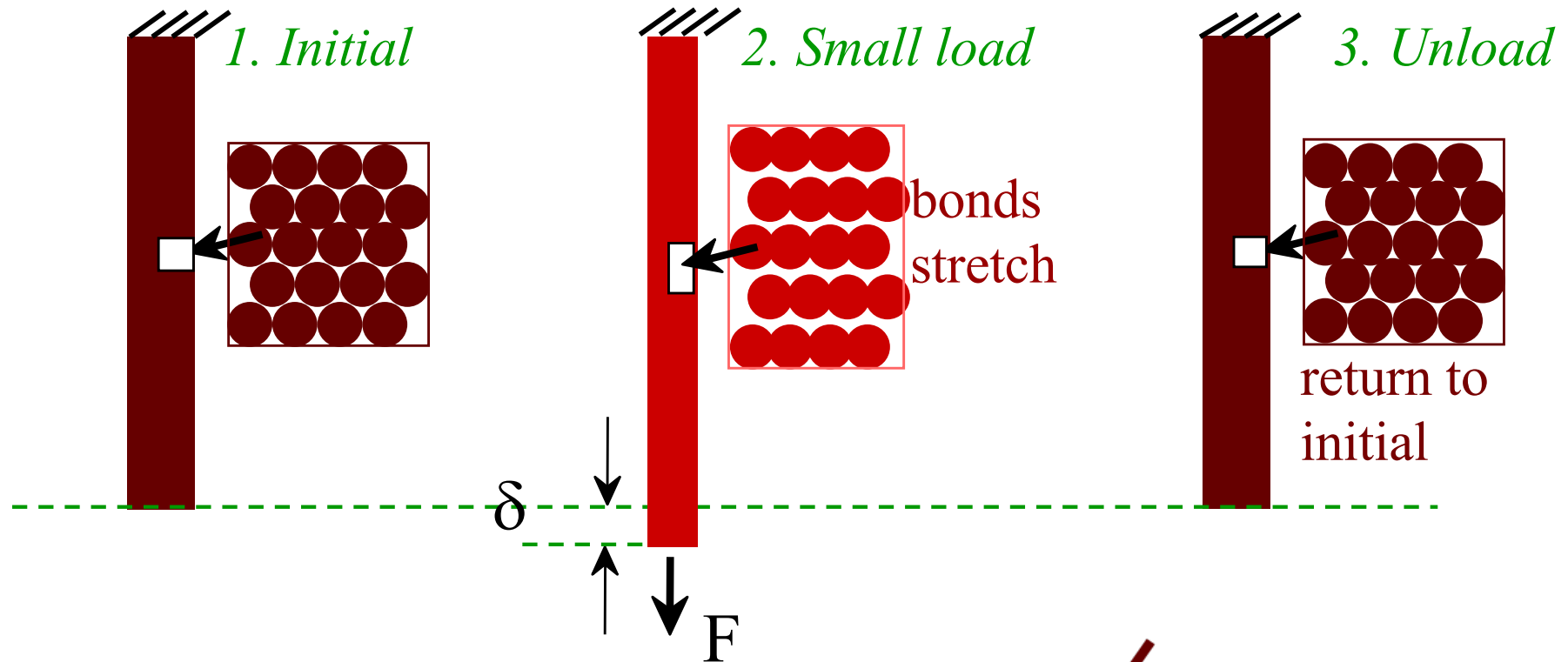
- Shear stress, τ :



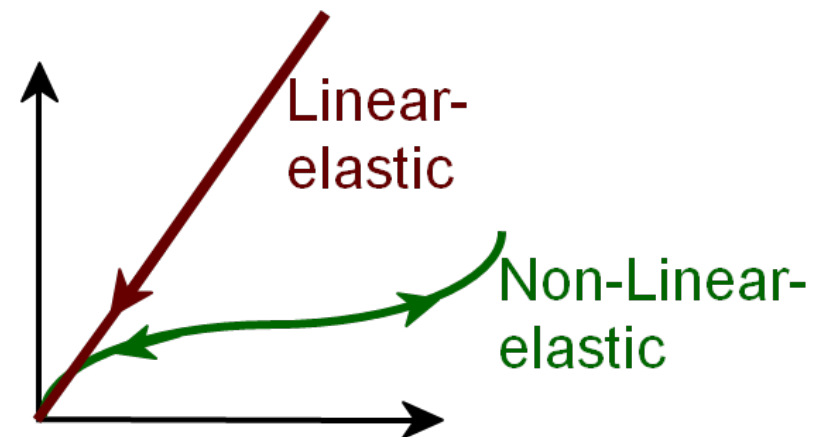
Stress has units: N/m^2 or lb/in^2



Elastic Deformation



Elastic means **reversible**!

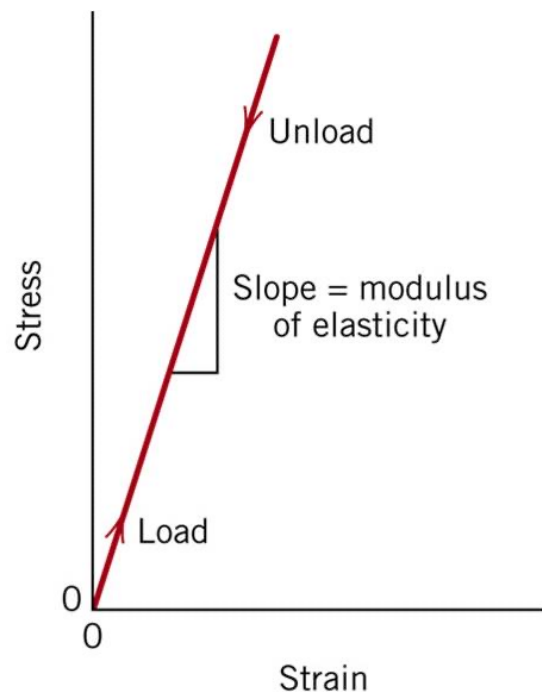




Stress-Strain Behaviour - *Elastic Response*

Initially, stress and strain are directly proportional to each other

- *Rationale*: atoms can be thought of as masses connected to each other through a network of **springs**.



According to Hooke's law, the extension of a spring, x , and the applied force, F , are related by the spring constant, k :

$$F = -kx$$

The constant of proportionality, *Young's modulus or modulus of elasticity*, is a measure of the material's stiffness.

Materials possessing high stiffness: W, Ta, Mo → slope

Materials possessing low stiffness: Al, Cu, Ag → slope



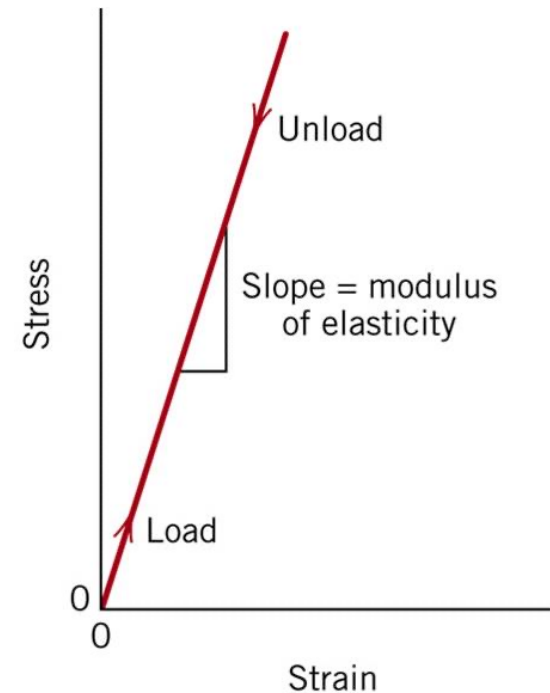
Stress-Strain Behaviour - *Elastic Response*

In the elastic region a material returns to its original dimensions when load released, and we can write

$$\sigma = E \varepsilon \quad \text{or} \quad \frac{F}{A_o} = E \frac{\Delta l}{l_o}$$

Example:

A steel wire with a cross sectional area of 0.55 mm^2 and length of 10 m is extended *elastically* 1.68 mm by a force of 17.24 N . What is the modulus of elasticity for this steel specimen?





Young's Modulus or Modulus of Elasticity

So E tells us how much something will stretch elastically when loaded, i.e. the **STIFFNESS** of that material

High E value - very stiff

Ceramics	300 GPa
Steel	207 GPa

Medium E value

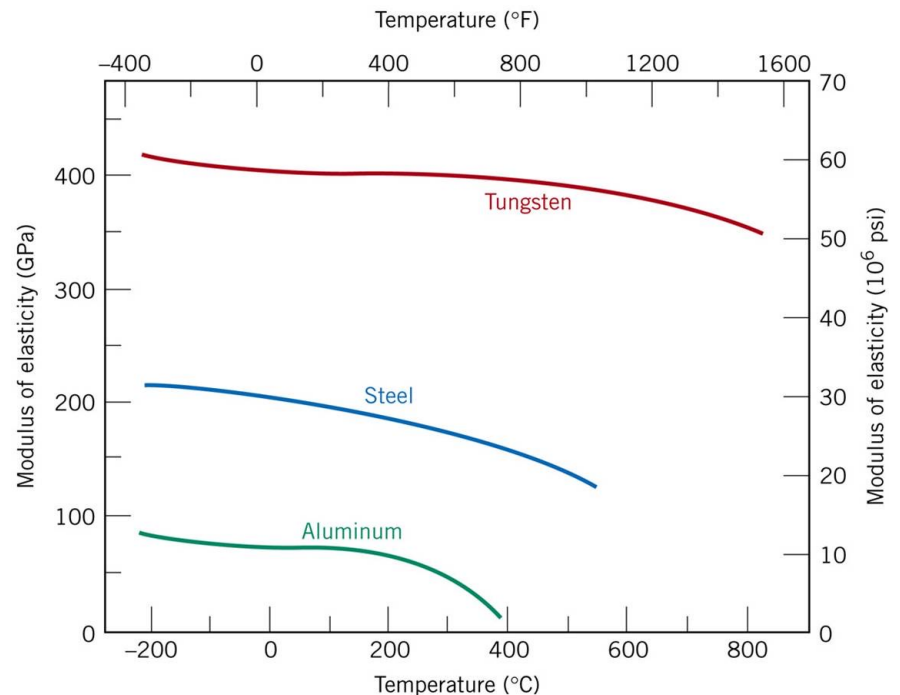
Copper	110 GPa
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Low E value: - (not stiff)

Plastics	3 GPa
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In general, the material's modulus (or stiffness) decreases with increasing temperature.

Can you think of why this happens?



Temperature dependence of elastic moduli

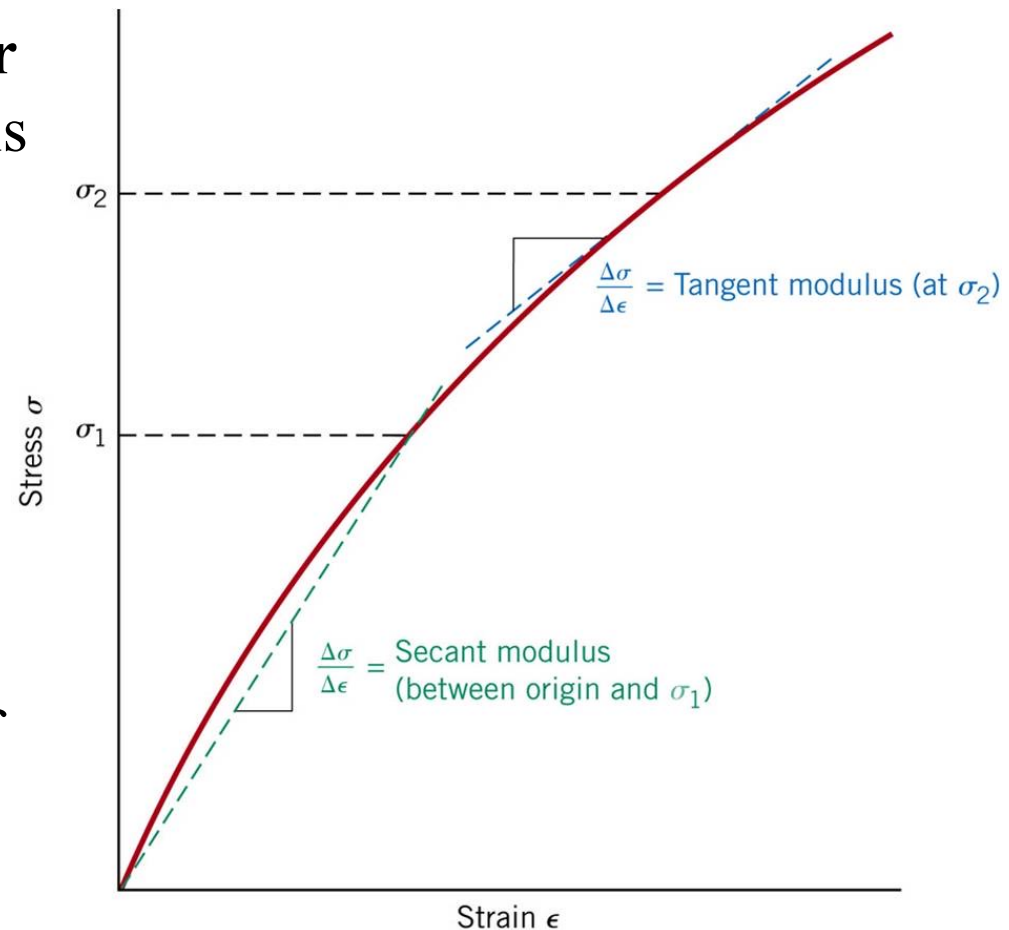


Tangent and Secant Modulus

- Some materials do not show **linear** elastic region; their elastic region is non-linear. Cast iron, concrete, some polymers.

In this case E is harder to define:

- We can use **Tangent modulus** which is slope of tangent at a particular stress level, or,
- Secant modulus** which is slope of the line joining origin with some specified stress level.

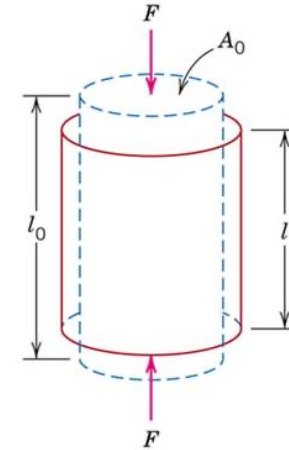




Compression, Shear and Torsion Tests

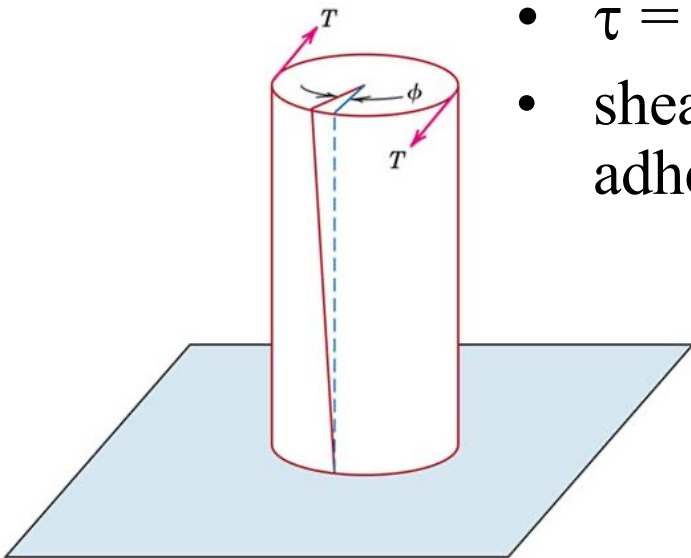
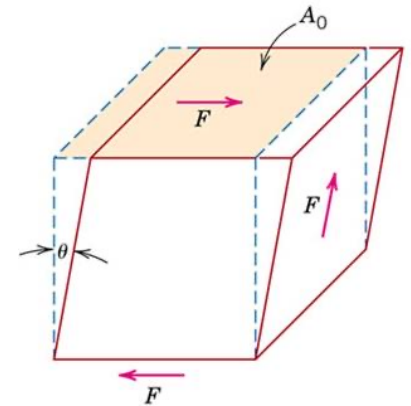
Compression

- by convention, stress and strain are negative
- used for measuring strength of brittle materials and for calculating forces required in manufacturing processes which involve compressive deformation



Shear

- shear stress is $\tau = F/A_0$ and γ (shear strain) is tangent of shear angle, θ
- $\tau = G \gamma$, G is **shear modulus**
- shear tests are often used to measure adhesive bonding, riveted joints etc



Torsion

- *torsion* is a variation of shear occurring in *machine axles, drive shafts and twist drills*
- $T = f(\tau)$ and $\gamma = f(\phi)$



Poisson's Ratio

- When pulled in tension (along Z), a sample gets **longer** and **thinner**, i.e., *a contraction in the width (X) and breadth (Y)*
- *if compressed gets fatter*
- **Poisson's ratio** defines how much strain occurs in the lateral directions (x & y) when strained in the (z) direction:

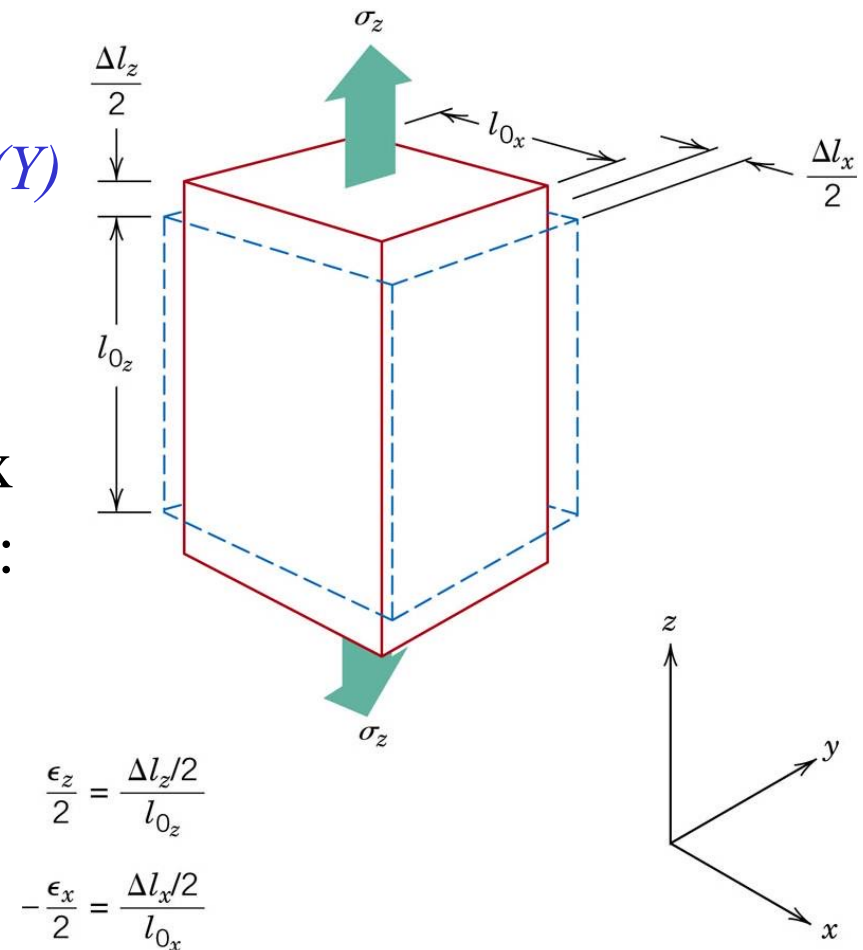
$$\nu = - \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$$\nu = - \frac{\epsilon_x}{\epsilon_z} = - \frac{\epsilon_y}{\epsilon_z}$$

- Typical values = 0.2 to 0.5

- For isotropic materials

$$E = 2G(1 + \nu)$$



Some materials are anisotropic so E & G vary with direction (e.g. composite materials and single crystals)



Elastic Deformation: Anelasticity

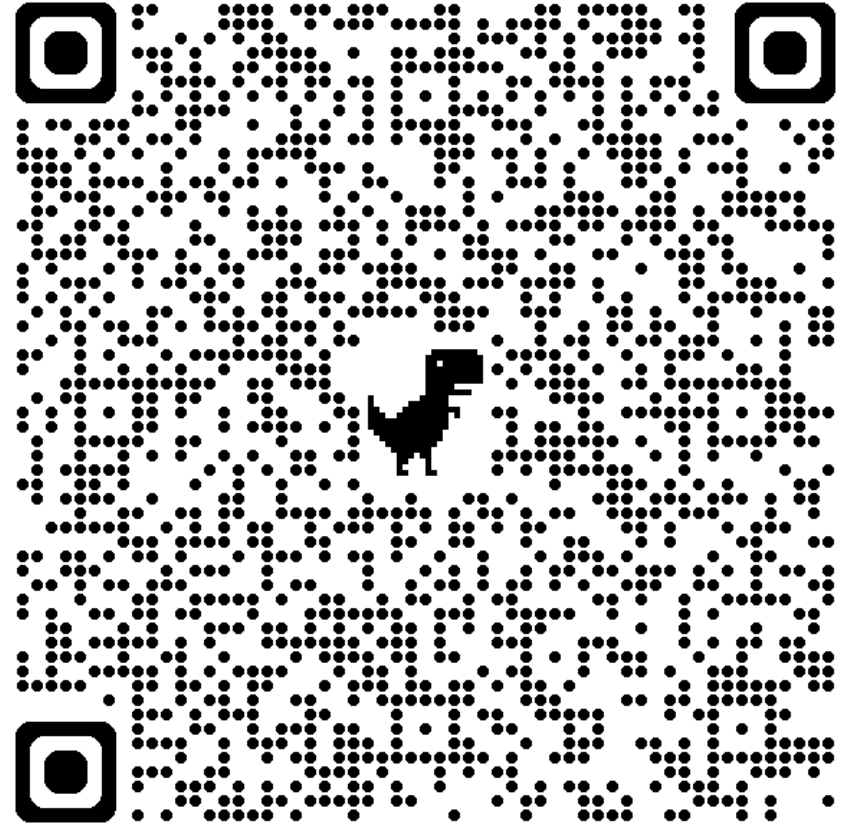
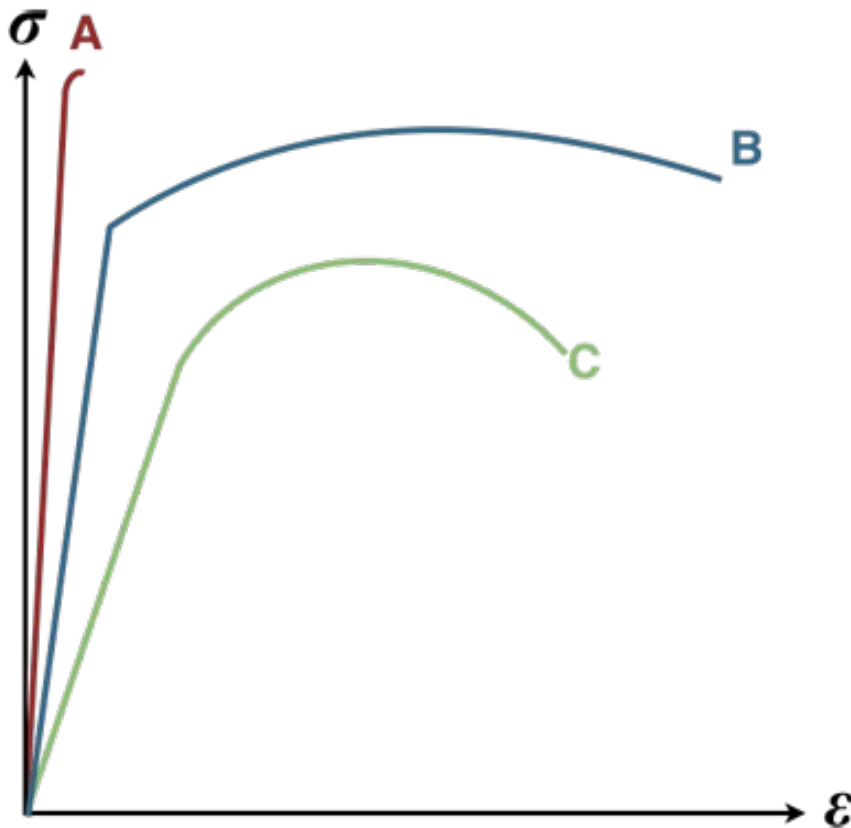
time dependence of elastic deformation

- So far we have assumed that elastic deformation is time (i.e. applied stress produces instantaneous elastic strain)
- In some materials, elastic strain may develop gradually after loading, and recovery may take time after unloading. This time-dependent, recoverable deformation is called **anelasticity**.
- The effect is normally small for **metals** but can be significant for (“viscoelastic behavior”).



Elastic Behaviour - Review

Which tensile engineering stress–strain curve represents the **least** stiff material?



Use the QR code shown in the classroom. This one won't work.



Next topic:

Plastic Deformation